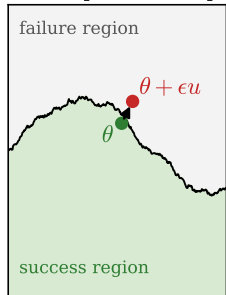


Twin simulators: repeat for many θ , random directions u , and scales ϵ

simulator-parameter space



same frozen
PPO policy

simulator A: θ



SUCCESS

same initial state, same policy

simulator B: $\theta + \epsilon u$



FAIL

$$f(\epsilon) = \Pr[S_A \neq S_B]$$

$$\propto \epsilon^\alpha$$

$$\alpha \approx 0.28$$

$$\text{smooth: } \alpha = 1$$

$\epsilon = 10^{-3}$ (0.1% mass change) still flips 5% of outcomes; halving f needs $2^{1/\alpha} \approx 12 \times$ better model accuracy